

Incident Investigation Report
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Incident Investigation Report

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Incident Information

Field	Value
Incident ID	SOC-2026-02-28-001
Incident Type	NetSupport Manager RAT Infection
Severity	High
Detection Source	SIEM Signature Alert
Analyst	Lakshmeesh R
Date Investigated	28-Feb-2026
Evidence Source	2026-02-28-traffic-analysis-exercise.pcap
Tool Used	Wireshark

Executive Summary

The Security Information and Event Management (SIEM) platform generated multiple alerts indicating suspected NetSupport Manager Remote Access Trojan (RAT) activity involving the external IP address **45.131.214.85** over TCP port **443**.

Packet capture analysis confirmed that an internal Windows workstation initiated outbound communication with this external host and subsequently exchanged HTTP POST traffic consistent with C2 check-in / beaconing behavior. Further investigation identified the affected workstation's IP address, MAC address, hostname, authenticated user account, and full user identity by correlating TCP/HTTP, LLMNR, Kerberos, and SAMR traffic.

The evidence indicates the workstation was compromised and communicated with malicious infrastructure, warranting immediate containment and follow-up forensic review.

Alert Details

Item	Value
Malware	NetSupport Manager RAT
External IP	45.131.214.85
Destination Port	TCP/443
Detection Time	2026-02-28 19:55 UTC (2026-03-01 01:25 IST in local packet captures)
Detection Source	SIEM Signature

Environment

Item	Value
Domain	EASYAS123.TECH
AD Domain	EASYAS123
Domain Controller	10.2.28.2
Gateway	10.2.28.1
Internal Network	10.2.28.0/24

Investigation Methodology

The packet capture was examined in Wireshark. The following protocols were analyzed to trace the incident from initial C2 communication back to host and user identity:

- TCP / HTTP — external command-and-control communication
- Statistics → Endpoints (Ethernet / IPv4) — host and MAC correlation
- LLMNR — hostname discovery
- Kerberos — authenticated user account
- SAMR — full user identity

The investigation began with the SIEM-flagged external IP and worked backward through the capture to positively identify the host, its owner, and the scope of exposure.

Investigation Findings

1. Infected Windows Client

Field	Value
IP Address	10.2.28.88

Evidence: Filtering on `ip.addr == 45.131.214.85` isolated all traffic between the internal network and the suspicious external host. The capture shows host **10.2.28.88** completing a TCP three-way handshake to **45.131.214.85** on port 443, followed by repeated HTTP POST requests to `http://45.131.214.85/fakeurl.htm` with `application/x-www-form-urlencoded` content — consistent with C2 beaconing. This traffic repeats at intervals (01:25, 01:26, 01:27), indicating periodic check-in behavior rather than a single connection. (See Figure 1.)

2. MAC Address

Field	Value
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MAC Address **00:19:d1:b2:4d:ad**

Evidence: Using Statistics → Endpoints, the IPv4 endpoint table confirms **10.2.28.88** as the internal host involved in the flagged traffic (*Figure 2*). Cross-referencing the Ethernet endpoint table for the same capture identifies the associated hardware address as **00:19:d1:b2:4d:ad** (*Figure 3*), which also appears as the source MAC in the LLMNR and outbound TCP frames reviewed elsewhere in this investigation.

3. Hostname

Field	Value
Hostname	DESKTOP-TEYQ2NR

Evidence: Filtering on `llmnr` surfaced repeated Link-Local Multicast Name Resolution queries originating from **10.2.28.88**, sourced from MAC **00:19:d1:b2:4d:ad**. The query name field resolves to **DESKTOP-TEYQ2NR**, confirming the infected host's Windows hostname. (*See Figure 4.*)

4. User Account

Field	Value
Username	brolf

Evidence: Filtering on `kerberos.CNameString` isolated Kerberos TGS-REP traffic between the Domain Controller (10.2.28.2) and the workstation (10.2.28.88) within the **EASYAS123.TECH** realm. The `CNameString` field in the ticket reply identifies the authenticated client principal as **brolf**. (*See Figure 5.*)

5. User Full Name

Field	Value
Full Name	Becka Rolf

Evidence: SAMR (Security Account Manager Remote) traffic between 10.2.28.2 and 10.2.28.88 was reviewed. The `QueryUserInfo` response (`Info21`) contains the account's `Full Name` field, which resolves to **Becka Rolf**, tying the Kerberos-authenticated account `brolf` to a real user identity. (*See Figure 6.*)

Indicators of Compromise (IOCs)

IOC Type	Value
External C2 IP	45.131.214.85
C2 URI	/fakeurl.htm
Internal Host	10.2.28.88
MAC Address	00:19:d1:b2:4d:ad
Hostname	DESKTOP-TEYQ2NR
Username	brolf
Full Name	Becka Rolf
Destination Port	TCP/443

Timeline

Time (UTC)	Time (IST, as captured)	Event
2026-02-28 19:55:11	01:25:11	Initial LLMNR queries observed from 10.2.28.88
2026-02-28 19:55:23	01:25:23	Kerberos TGS-REP / SAMR traffic confirms user brolf (Becka Rolf)
2026-02-28 19:55:51	01:25:51	TCP SYN from 10.2.28.88 to 45.131.214.85:443 (C2 connection established)
2026-02-28 19:55:51	01:25:51	First HTTP POST to /fakeurl.htm — C2 check-in begins
2026-02-28 19:56:52	01:26:52	Repeat HTTP POST beacon
2026-02-28 19:57:52	01:27:52	Repeat HTTP POST beacon

Evidence Summary

Evidence	Protocol / View	Figure
Malicious C2 traffic	TCP / HTTP	Figure 1
Host identification	Statistics → Endpoints (IPv4)	Figure 2
MAC identification	Statistics → Endpoints (Ethernet)	Figure 3
Hostname discovery	LLMNR	Figure 4
User authentication	Kerberos	Figure 5
Full user identity	SAMR	Figure 6

MITRE ATT&CK Mapping

Technique	ID	Notes
Remote Access Software	T1219	NetSupport Manager RAT used as remote access tooling
Application Layer Protocol	T1071	HTTP used for C2 communication over port 443
Non-Standard Port / Encoding	T1071.001 (sub)	Repeated form-urlencoded POSTs to a single URI resemble a beaconing pattern

Note on T1105 (Ingress Tool Transfer): The capture shows repeated HTTP POST requests (client → server) rather than a file download (server → client) of meaningful size. This is consistent with C2 check-in/exfiltration traffic, not a confirmed payload transfer. T1105 is therefore **not included** in this mapping; it should only be added if a subsequent memory or disk forensic review confirms a dropped or downloaded second-stage tool.

Impact Assessment

Host **10.2.28.88 (DESKTOP-TEYQ2NR)**, used by domain user **brolf (Becka Rolf)**, established outbound communication with confirmed malicious infrastructure at **45.131.214.85** and exchanged multiple

HTTP POST beacons over roughly two minutes of captured traffic. Because the same host also completed normal LLMNR, Kerberos, and SAMR exchanges with the domain controller (10.2.28.2) in the surrounding window, the infection was active on a domain-joined endpoint with a live authenticated session — not an isolated or air-gapped machine.

The practical risk here is twofold:

- **Credential exposure risk:** the account **brolf** authenticated to the domain (**EASYAS123.TECH**) while the host was compromised. If the RAT has keylogging or credential-harvesting capability (common in NetSupport Manager RAT deployments), the domain account should be treated as potentially compromised, not just the workstation.
- **Lateral movement risk:** repeated beaconing over ~2 minutes suggests the implant was still actively communicating with C2 at the time of capture, which typically precedes attacker-directed follow-on activity (reconnaissance, credential dumping, or lateral movement) rather than being the end state of the intrusion.

No evidence in this capture confirms data exfiltration volume, lateral movement to other hosts, or privilege escalation — those require endpoint (EDR/memory) forensics beyond what this pcap can show.

Recommendations

Immediate

- **Isolate the endpoint:** quarantine **10.2.28.88** via EDR network isolation or switch-port shutdown (not just a firewall rule) to preserve volatile memory for forensic capture while cutting off C2 access.
- **Block the C2 indicator:** block **45.131.214.85** and the URI pattern `/fakeurl.htm` at the perimeter firewall and web proxy.
- **Contain the account:** disable **brolf** and force a credential reset once the endpoint is isolated; treat the account as potentially compromised, not just the device.
- **Forensic capture:** collect a memory image and disk image from **10.2.28.88** before remediation to determine initial access vector and any additional dropped tooling.

Long-Term

- Hunt enterprise-wide for connections to **45.131.214.85** and the `/fakeurl.htm` URI pattern.
 - Review outbound HTTP/HTTPS traffic for other hosts exhibiting the same POST/beacon interval pattern.
 - Deploy or tune EDR detection for NetSupport Manager RAT process and network signatures.
 - Strengthen SIEM correlation rules to alert on LLMNR + Kerberos + SAMR + unusual outbound HTTP occurring in a short window on the same host, which is what made this identification possible.
 - Conduct a broader credential hygiene review if the RAT is confirmed to have keylogging capability.
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Conclusion

The investigation confirmed that Windows workstation **10.2.28.88 (DESKTOP-TEYQ2NR)**, MAC **00:19:d1:b2:4d:ad**, used by domain account **brolf (Becka Rolf)**, established outbound C2 communication with **45.131.214.85** over TCP port 443, exchanging repeated HTTP POST beacons to `/fakeurl.htm`. Correlating LLMNR, Kerberos, SAMR,

Ethernet/IPv4 endpoint data, and the C2 traffic itself provided sufficient evidence to positively identify the affected asset and its user, supporting a finding of compromise consistent with a NetSupport Manager RAT infection. The recommendations above should be actioned before the host is returned to service.

Appendix — Evidence Screenshots

Figure 1. Filtered view (ip.addr == 45.131.214.85) showing the TCP handshake from 10.2.28.88 to 45.131.214.85:443, followed by repeated HTTP POST requests to /fakeurl.htm — the initial identification of the C2 channel.

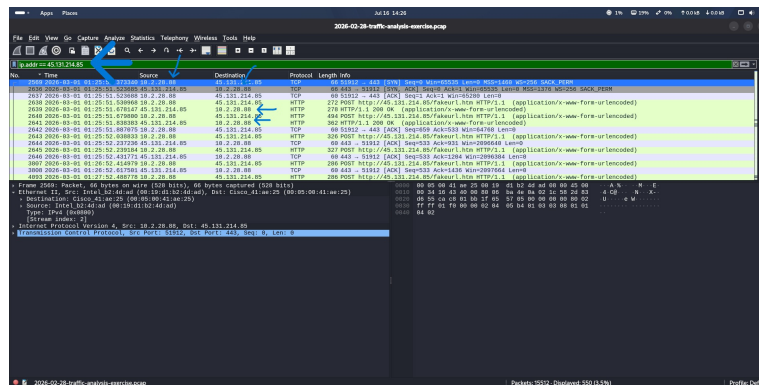


Figure 1

Figure 2. Wireshark Statistics → Endpoints (IPv4) view, confirming 10.2.28.88 as the internal host involved in the flagged session.

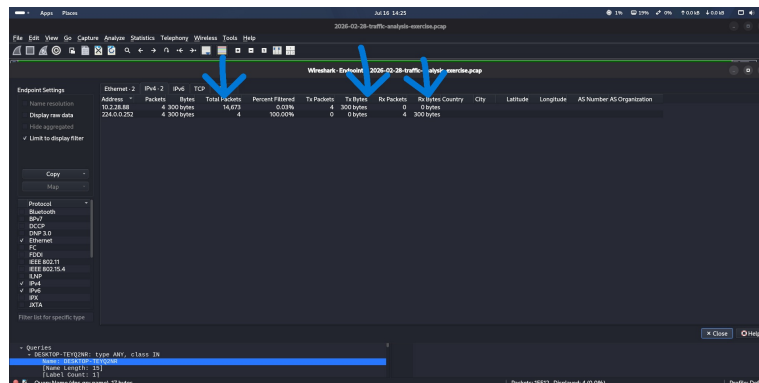


Figure 2

Figure 3. Wireshark Statistics → Endpoints (Ethernet) view, identifying MAC address 00:19:d1:b2:4d:ad as associated with the infected host.

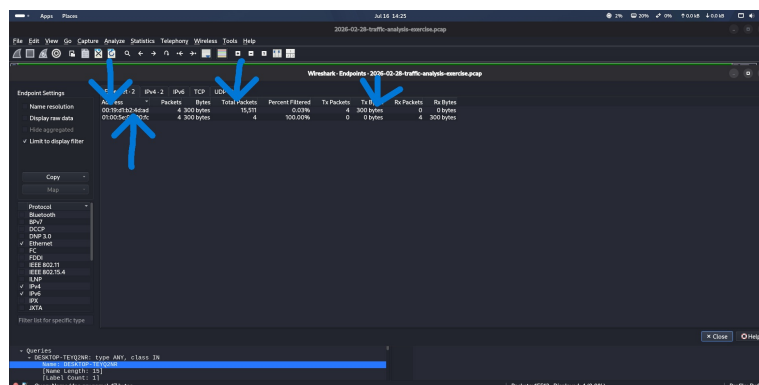


Figure 3

Figure 4. LLMNR query traffic (llmnr filter) from 10.2.28.88 resolving hostname DESKTOP-TEYQ2NR.

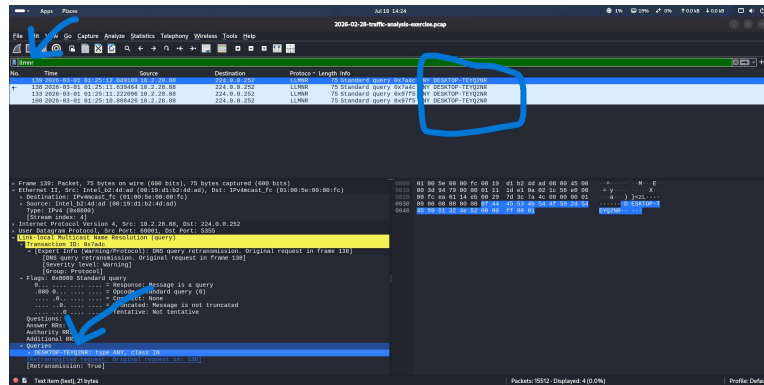


Figure 4

Figure 5. Kerberos TGS-REP traffic (kerberos.CNameString filter) identifying authenticated client principal brolf.

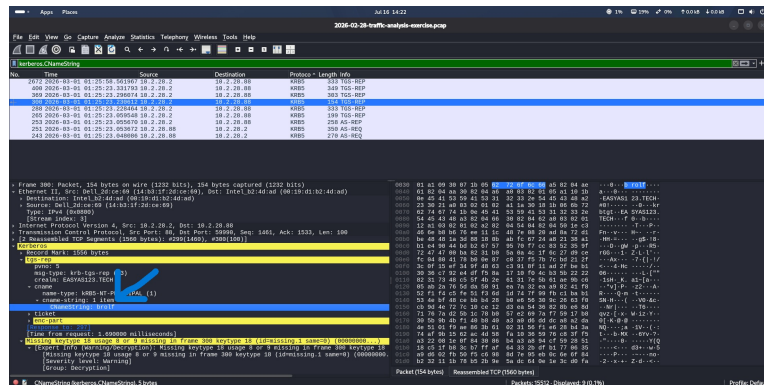


Figure 5

Figure 6. SAMR QueryUserInfo response (Info21) resolving account brolf to full name "Becka Rolf".

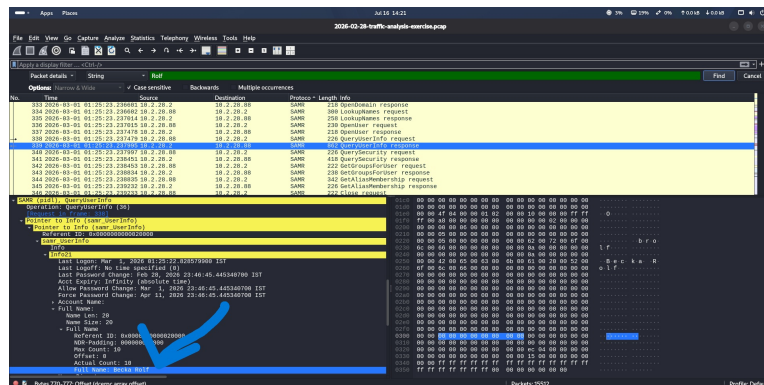


Figure 6

Report prepared by Lakshmeesh R as part of a SOC investigation portfolio exercise.